

Normalisation

Port Tours Available

Port Code AUHOB
Port Name Hobart
Country Code AU
Country Name Australia

Port Temperature

Month	Average High	Average Low
Jan	22	13
Feb	22	13
Mar	20	12
Apr	18	10
May	15	8
Jun	12	6
July	12	6
Aug	14	6
Sep	16	7
Oct	17	8
Nov	19	10
Dec	21	11

Tours Available

Tour Number	Tour Name	Tour Description	Hours Required	Tour Cost Per Person	Wheel Chair Access	Tour Availability	Start Time
1	Hobart Shopping	Shop at the Cat & Fiddle Arcade for a full range of Tasmanian treats (bus transfer only)	2.5	15	Yes	Weekdays	11:00
2	Port Arthur	Visit the world heritage listed 19th century penal settlement and open-air museum via local ferry. Includes lunch and entry	8	270	No	Daily	09:00
3	MONA	Visit the Museum of Old and New Art (MONA) - includes ferry ride to the museum and entry	4	70	Yes	Daily	13:00
4	Salamanca Market	Shop in the bohemian market stalls of the Salamanca market selling the best local produce and goods (bus transfer only)	3	15	Yes	Saturday	11:00

UNF

PORT (port_code, port_name, country_code, country_name (temp_month, temp_high, temp_low)
 (tour_number, tour_name, tour_description, tour_hours, tour_cost_pp, tour_wchair_access, tour_when,
 tour_starttime)

1NF

PORT (port_code, port_name, country_code, country_name)

PORT_TEMP (port_code, temp_month, temp_high, temp_low)

TOUR (port_code, tour_number, tour_name, tour_description, tour_hours, tour_cost_pp,
 tour_wchair_access, tour_when, tour_starttime)

Partial Dependency:

No Partial Dependency

2NF

PORT (port_code, port_name, country_code, country_name)

PORT_TEMP (port_code, temp_month, temp_high, temp_low)

TOUR (port_code, tour_number, tour_name, tour_description, tour_hours, tour_cost_pp, tour_wchair_access, tour_when, tour_starttime)

Transitive Dependency:

country_code -> country_name

3NF

COUNTRY (country_code, country_name)

PORT (port_code, port_name, country_code)

PORT_TEMP (port_code, temp_month, temp_high, temp_low)

TOUR (port_code, tour_number, tour_name, tour_description, tour_hours, tour_cost_pp, tour_wchair_access, tour_when, tour_starttime)

Full Dependency:

country_code -> country_name

port_code -> port_name, country_code

port_code, temp_month -> temp_high, temp_low

port_code, tour_number -> tour_name, tour_description, tour_hours, tour_cost_pp, tour_wchair_access, tour_when, tour_starttime

Port - Tour Participant Report

Port Code AUHOB
Tour Number 3
Tour Name MONA
Tour Date 25th March 2022
Tour Start Time 13:00

Generated: 24th March 2022

Passenger ID	Passenger Name	Principal Spoken Language	Cruise ID	Cruise Name	Payment Received
10210	Gilberto Bwy	English	1024	Sydney to Hobart	Yes
10211	Fransyne Bwy	English	1024	Sydney to Hobart	Yes
10029	Friedrick Geist	German	1095	New Zealand Sea Escape	No
10789	Brier Kilgour	English	1095	New Zealand Sea Escape	No
10301	Shandra Lindblom	French	1024	Sydney to Hobart	Yes
10302	Pierre Lindblom	French	1024	Sydney to Hobart	Yes
10303	Michael Lindblom	French	1024	Sydney to Hobart	Yes
13456	Myriam Stirley	English	1095	New Zealand Sea Escape	Yes

.... Only partial data shown

UNF

TOUROFFER (port_code, tour_number, tour_name, tour_date, tour_starttime (passenger_id, passenger_fname, passenger_lname, passenger_language, cruise_id, cruise_name, tour_paid))

Note passenger name must be decomposed - Ass1A indicated it was made up of first and last names (see Week 6 Applied Notes page 7)

INF

TOUROFFER (port_code, tour_number, tour_name, touroffer_date, tour_starttime)

PARTICIPANT (port_code, tour_number, touroffer_date, passenger_id, passenger_fname, passenger_lname, passenger_language, cruise_id, cruise_name, tour_paid)

PARTIAL:

port_code, tour_number -> tour_name, tour_starttime

passenger_id -> passenger_fname, passenger_lname, passenger_language

2NF

TOUR (port_code, tour_number, tour_name, tour_starttime)

TOUROFFER (port_code, tour_number, touroffer_date)

PARTICIPANT (port_code, tour_number, touroffer_date, passenger_id, cruise_id, cruise_name, tour_paid)

PASSENGER (passenger_id, passenger_fname, passenger_lname, passenger_language)

TRANSITIVE

cruise_id -> cruise_name

3NF

TOUR (port_code, tour_number, tour_name, tour_starttime)

TOUROFFER (port_code, tour_number, touroffer_date)

PARTICIPANT (port_code, tour_number, touroffer_date, passenger_id, cruise_id, tour_paid)

PASSENGER (passenger_id, passenger_fname, passenger_lname, passenger_language)

CRUISE (cruise_id, cruise_name)

Full Dependency:

port_code, tour_number -> tour_name, tour_starttime

port_code, tour_number, touroffer_date, passenger_id -> cruise_id, tour_paid

passenger_id -> passenger_fname, passenger_lname, passenger_language

cruise_id -> cruise_name

ATTRIBUTE SYNTHESIS

1. COUNTRY (country_code, country_name)

2. PORT (port_code, port_name, country_code)

3. PORT_TEMP (port_code, temp_month, temp_high, temp_low)

4. TOUR (port_code, tour_number, tour_name, tour_description, tour_hours, tour_cost_pp, tour_wchair_access, tour_when, tour_starttime)

5. TOUR (port_code, tour_number, tour_name, tour_starttime)

6. TOUROFFER (port_code, tour_number, touroffer_date)

7. PARTICIPANT (port_code, tour_number, touroffer_date, passenger_id, cruise_id, tour_paid)

8. PASSENGER (passenger_id, passenger_fname, passenger_lname, passenger_language)

9. CRUISE (cruise_id, cruise_name)

Join 4 & 5

FINAL 3NF

COUNTRY (country_code, country_name)

PORT (port_code, port_name, country_code)

PORT_TEMP (port_code, temp_month, temp_high, temp_low)

TOUR (port_code, tour_number, tour_name, tour_description, tour_hours, tour_cost_pp, tour_wchair_access, tour_when, tour_starttime)

TOUROFFER (port_code, tour_number, touroffer_date)

PARTICIPANT (port_code, tour_number, touroffer_date, passenger_id, cruise_id, tour_paid)

PASSENGER (passenger_id, passenger_fname, passenger_lname, passenger_language)

CRUISE (cruise_id, cruise_name)